System Description of ShanHe-Huawei_OceanStor_A800_1host-1A800

1. Basic Information

1.1 System name of the submission

ShanHe-Huawei_OceanStor_A800_1host-1A800

1.2 Submitter name

JNIST

1.3 Submission date

July-10-2025

1.4 Version of the benchmark

MLPerf Storage V2.0

1.5 Solution type of the submission

Proprietary - OceanStorOS

1.6 Submission division (OPEN or CLOSED)

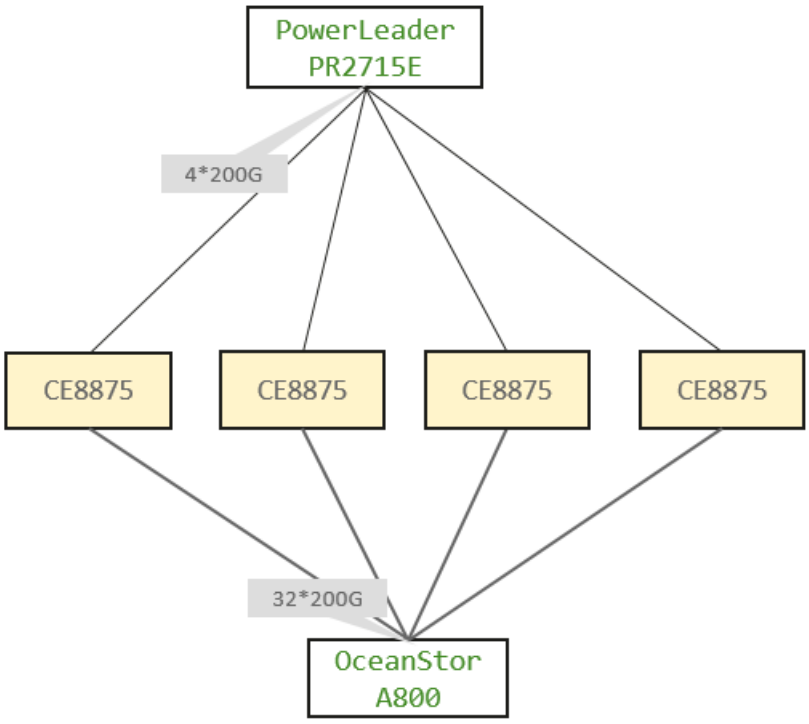
CLOSED

1.7 Power Requirements

System component	Power supply unit	Nameplate rated power	Design Power
A800 Appliance 1	Power supply 1	2000W	6000W
	Power supply 2	2000W	
	Power supply 3	2000W	
	Power supply 4	2000W	
	Power supply 5	2000W	
	Power supply 6	2000W	
	Power supply 7	2000W	
	Power supply 8	2000W	
Totals		16000W	6000W

1.8 System Topology

The test environment consists of one host and one Huawei OceanStor A800, connected to 4 ethernet RoCE switches.



1.9 Rack Units Requirements

The Storage system consumes 8U space.

2. Hardware

2.1 Host

The host node is a commercial servers with 2 Intel(R) Xeon(R) Platinum 8458P Processors. The Clients communicate with the storage through RoCE. The hardware details are as follows:

Component	Description	Count
CPU	Intel(R) Xeon(R) Platinum 8458P	2
Memory	DDR5 4800MHz 32GB	16
Network Card	Mellanox Technologies MT2892 Family [ConnectX-6 Dx]	4

2.2 Storage

The Huawei OceanStor A800 is a new generation of high-performance AI storage specifically designed for AI. It adopts an innovative control-data separation architecture, delivering ultimate performance to accelerate the training and inference of large models.
The Huawei OceanStor A800 high-performance AI storage supports flexible scaling, with a single cluster capable of supporting 512 controllers for scale-out and EB-level storage capacity expansion, meeting the smooth evolution needs of trillion/quadrillion-parameter multimodal large models. Additionally, the storage natively supports new data paradigms such as vectors, with a built-in high-performance vector retrieval engine that accelerates the retrieval of large-capacity vector knowledge bases, reduces inference "hallucinations," and provides KV Cache multi-level caching capabilities to reduce inference latency and improve the cost-effectiveness of inference. This storage system meets the full-process data processing needs of AI training and inference.



The hardware details of one appliance are as follows:

Component	Description	Count
CPU	HiSilicon Kunpeng-920	8
Memory	DDR5 32GB	64
Network Card	2*200G RoCE	16
Drive	NVMe SSD 7.68TB	64

2.3 Switch

Four Huawei CloudEngine 8875 switches are used. The host and storage are connected to it.

3 Software

3.1 Host

The configuration of the host software is shown as follows:

Software	Description
OS	openEuler release 22.03 (LTS-SP4)
Kernel	5.10.0-216.0.0.115.oe2203sp4.x86_64
Mellanox OFED version	MLNX_OFED_LINUX-5.8-1.1.2.1

3.2 Storage

The software deployed on the storage is OceanStor A800 V700R001C10.